
Sources
MUSAN: train, val, in-domain test (70/10/20 % by file)
LibriSpeech, GTZAN, ESC-50: unseen-source test
Mixing
10-s clips of 1-4 s events
SMR −5 to 15 dB
exact 10-ms labels
bg
sp
mu
sp+mu
CRNN
log-mel, 64 bands
3 conv + 1 GRU layer
86 K parameters
4 classes per frame
Training loss
cross-entropy
weighted CE
focal (γ = 2)
label smoothing (ε = 0.1)
× 3 seeds each
Calibration
fitted on validation set after training:
temperature (T)
vector (w, b)
Evaluation
macro-F1, class F1
ECE, classwise-ECE
fixed thresholds
clip-level bootstrap